5 Fig. 5.1 shows a 1.6 m long solenoid with 400 turns and a cross-sectional diameter of 0.040 m. A coil Y, with 80 turns, is wound tightly around the centre region of the solenoid.

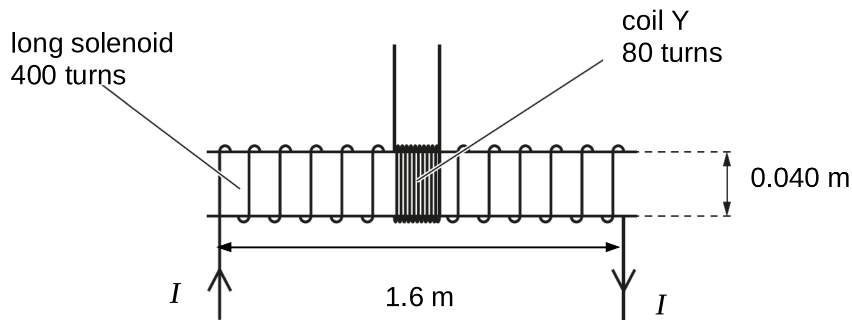

Fig. 5.1

(a) Show that, for a current I of 3.8 A in the solenoid, the magnetic flux linkage of coil Y is 1.2×10^{-4} Wb.

[2]

(b) The current I in the solenoid in **(a)** is reversed in 0.30 s.

Calculate the mean e.m.f. induced in coil Y.

e.m.f. = ... V [2]

(c) The current I in the solenoid varies with time t as shown in Fig. 5.2.

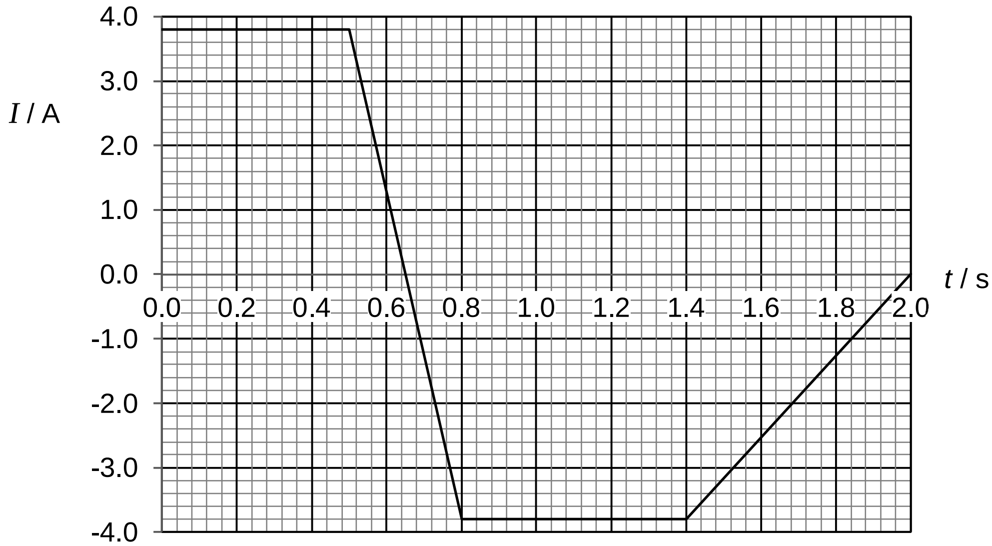

Fig. 5.2

Use your answer to **(b)** to sketch, on Fig. 5.3, the variation with time t of the e.m.f. E induced in coil Y for time t = 0 to time t = 2.0 s.

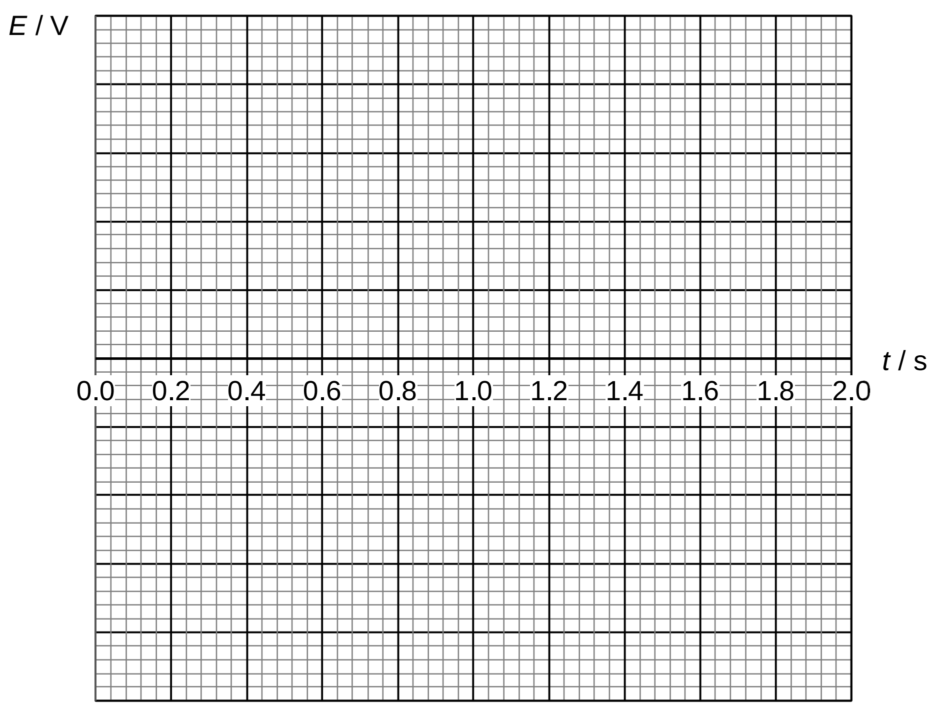

Fig. 5.3

[3]

(d) State and explain the effect on your answer in **(b)** if there is now an iron core placed in the solenoid.

....

....

.... [1]

6 **(a)** By reference to the photoelectric effect, explain